

# GuardSynth: Formal Specification and Verification of Behavioral Constraints for CoC-Based VLM Learning

## Abstract

Chain of Cognition (CoC) connects scene understanding with action reasoning for vision-language model (VLM) learning. However, learning from CoC alone can yield poor action selection when constraint applicability and release conditions remain unspecified. We present GuardSynth, a framework that augments CoC with explicit behavioral constraints. We define their scope and constituent elements and specify applicability, prohibited actions, persistence and release conditions, timing, and evidence using Evidence-Bound Lifecycle Contracts (EBLC). We then verify consistency and designated properties within a restricted logical model, generate Controlled Natural Language (CNL) corresponding to specification fields, and add it to CoC training inputs. We compare Qwen3-VL-2B-Instruct under matched LoRA training budgets across three seeds on a controlled synthetic visual-temporal task. Mean action-selection accuracy is 49.86% with CoC alone, 96.81% with our existing natural-language constraints, and 98.06% with EBLC-derived CNL; corresponding constraint-violation rates are 19.17%, 1.39%, and 0.83%. These results demonstrate a systematic path from behavioral constraint specification and verification to CoC augmentation, alongside improved action selection with explicit constraints. The findings concern a controlled task with constraints supplied during both training and evaluation; they establish neither a causal performance benefit of verification itself nor real-road safety.